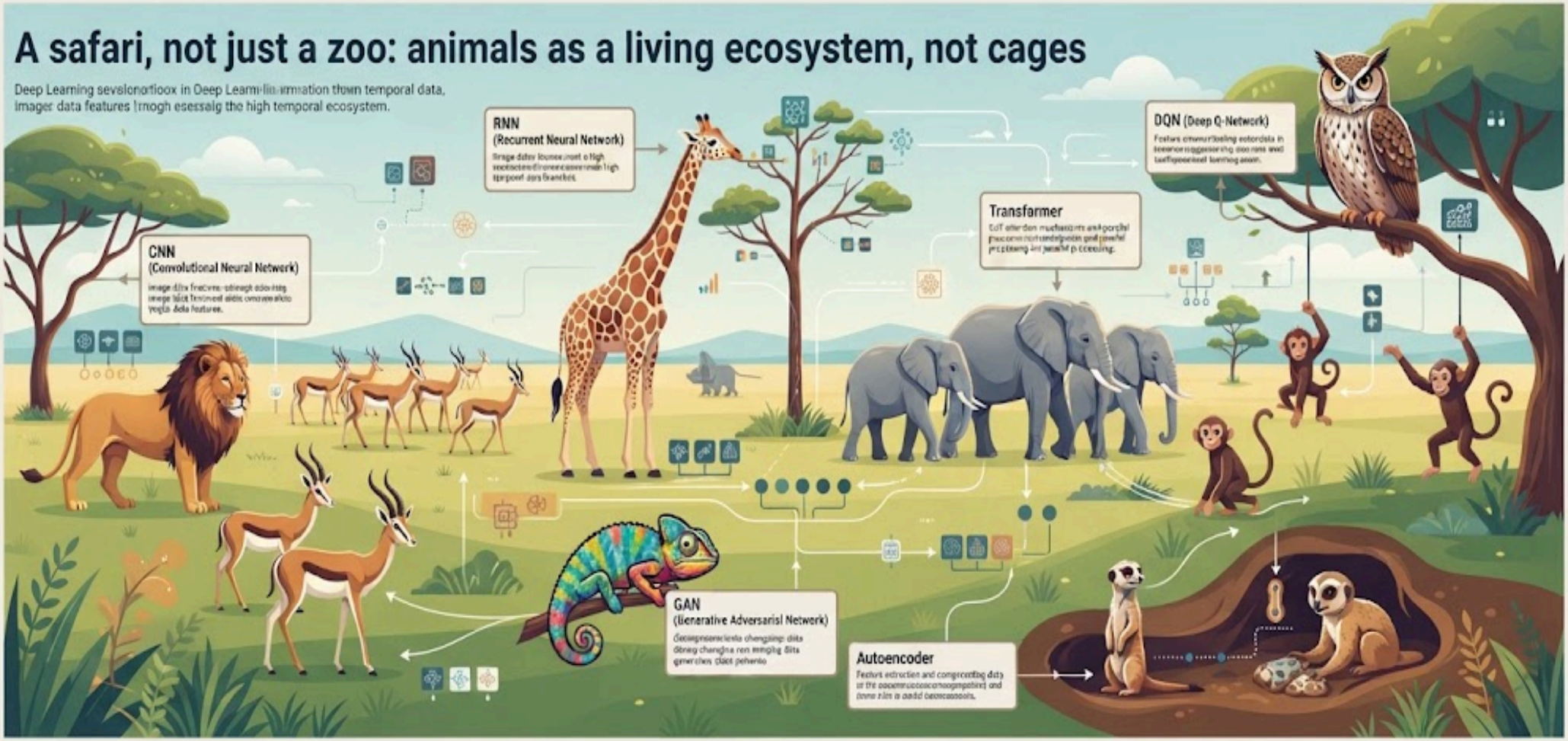
A03: The Neural Network Zoo

Team 3 | ITAI 2376 - Deep Learning
Submitter: Michael Rivas

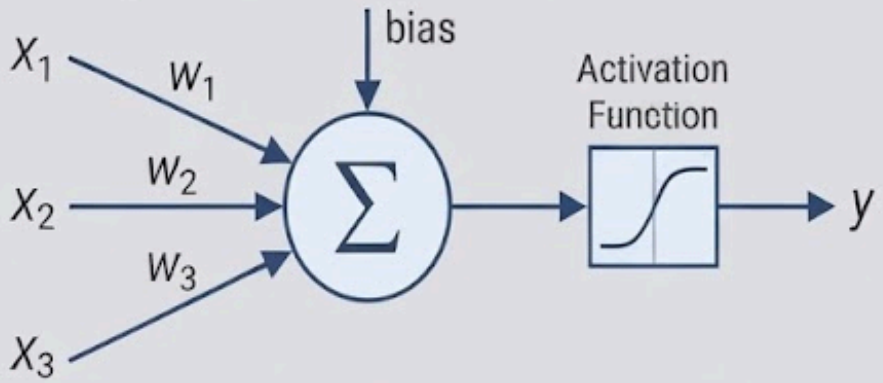


A03 Neural Network Zoo | Team 3 | ITAI 2376 | 1

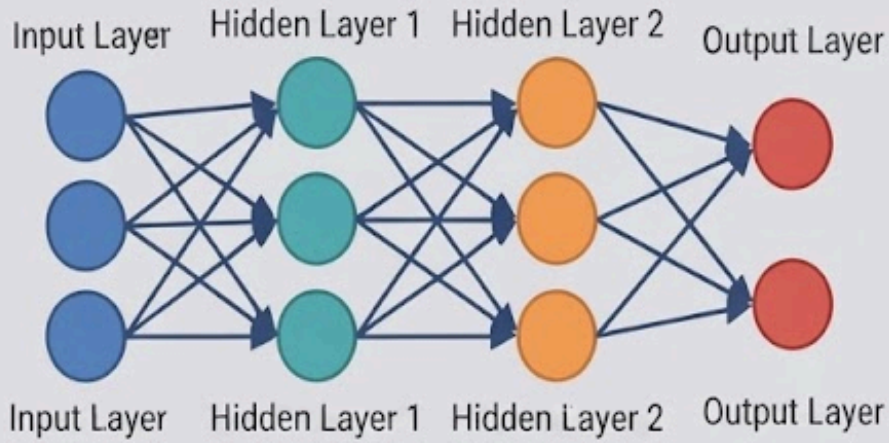
1. What is a neural network?

- Neural networks are computing systems inspired by biological brains.
- They process inputs by applying weights and adding a bias, followed by a non-linear activation function.
- Activation functions (e.g., ReLU) determine which signals are passed forward.

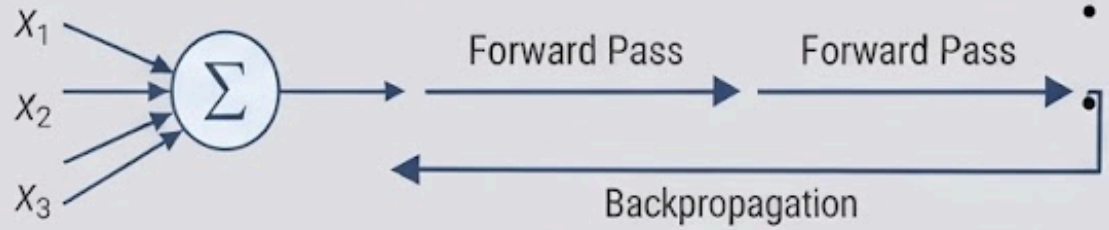
1. Neurons weigh inputs, add a bias, and fire through an activation function



2. Layers: input -> hidden layers -> output



3. Learning process: forward pass and backpropagation



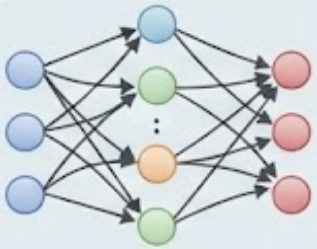
- Forward Pass: Data propagates from input to output layer to generate a prediction.
- Loss Calculation: Predicted output is compared to actual target value to calculate error.
- Backpropagation: The network calculates error gradients and adjusts weights backwards to minimize future loss.

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2. The Zoo / Safari Map

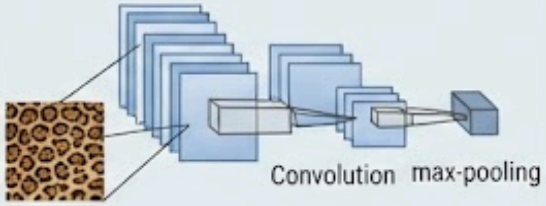


Ostrich - FNN / MLP: straight ahead, no loops



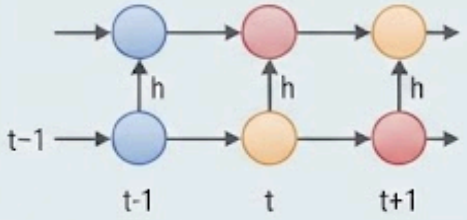


Leopard - CNN: local patterns, like rosettes, become objects



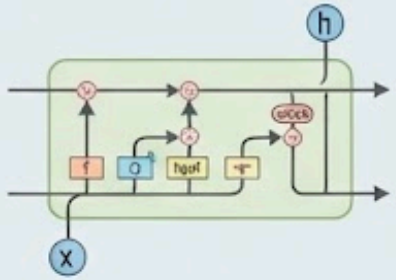


Hyena - RNN: calls relayed step by step, hidden state carried forward



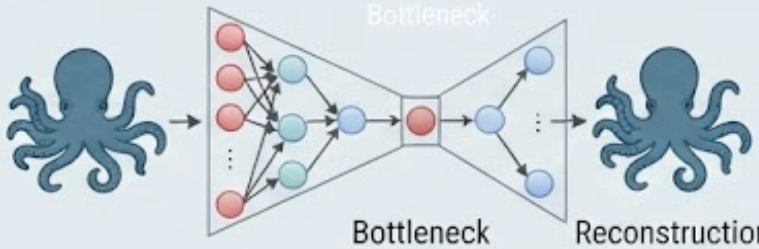


Parrot - LSTM: gates remember and forget over long sequences



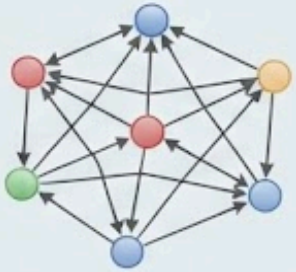


Octopus - Autoencoder: squeeze through a bottleneck, then rebuild



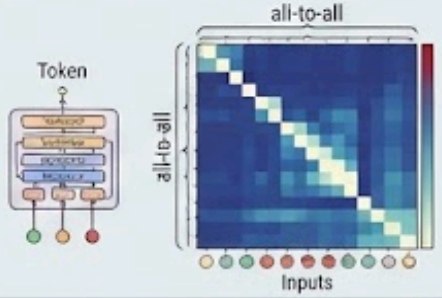


Spider - GNN: messages across a web of nodes





Honeybee - Transformer: every token can attend to every other token



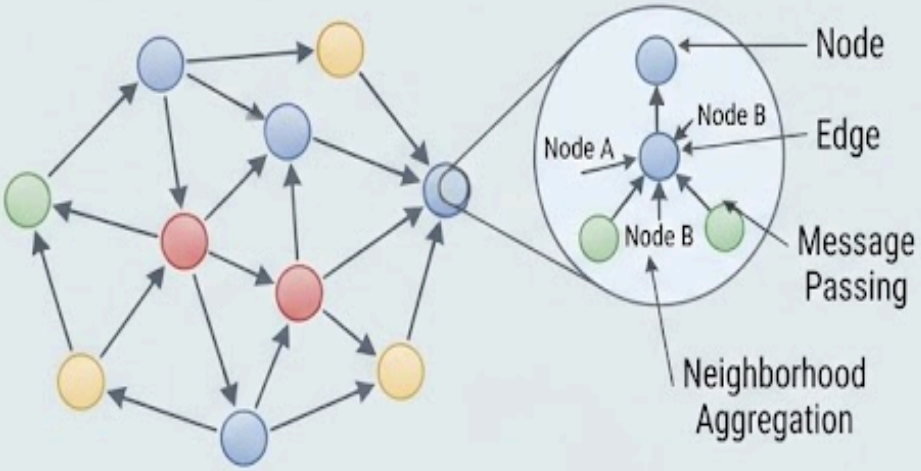
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3. Interactive Group Activity

Spider / GNN

Research & Structure: Understand Graph Neural Networks.
Focus on message passing across a web of nodes.

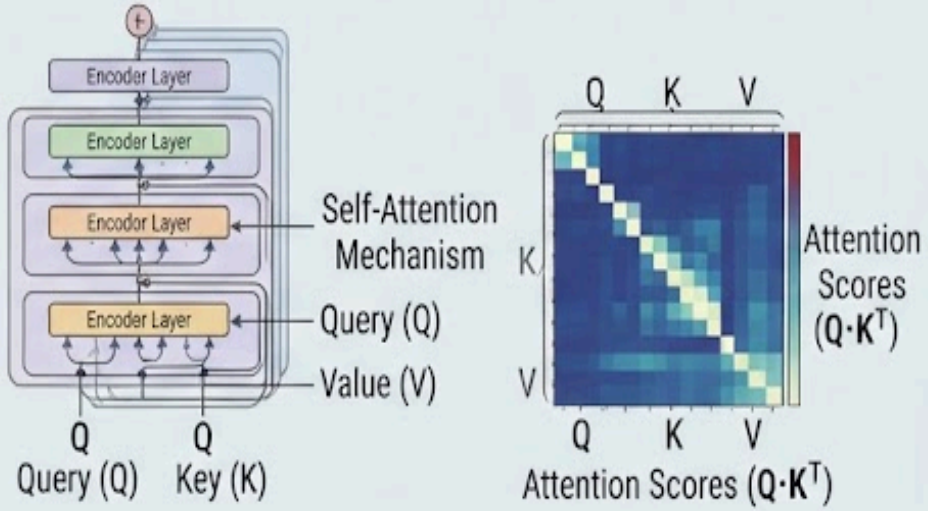


Live Notebook Cell Iteration

```
def update_node(node, neighbors):
    messages = [n.message for n in neighbors]
    node.hidden_state = aggregate(messages)
    return node.hidden_state
```

Honeybee / Transformer

Research & Structure: Understand Transformer Networks.
Focus on all-to-all token attention.



Live Notebook Cell Iteration

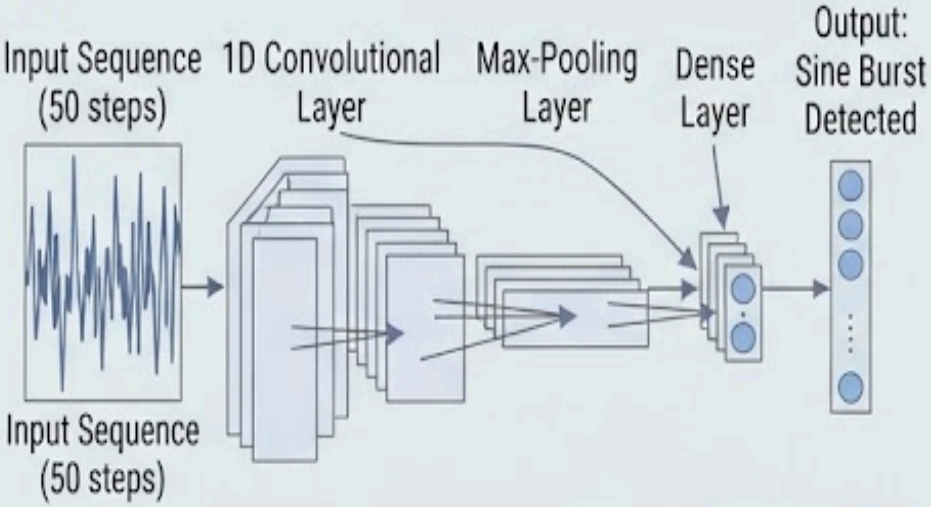
```
def scaled_dot_product_attention(Q, K, V):
    scores = matmul(Q, K.T) / sqrt(dk)
    attention = softmax(scores)
    return matmul(attention, V)
```

A03 Neural Network Zoo | Team 3 | ITAI 2376 | 1

4. A working program: CNN vs LSTM

Task & Data: Task: find a short sine burst in 50-step noisy sequences | **Data:** 1,000 sequences, 800 train / 200 test, 15 epochs, Adam

Leopard 1D CNN

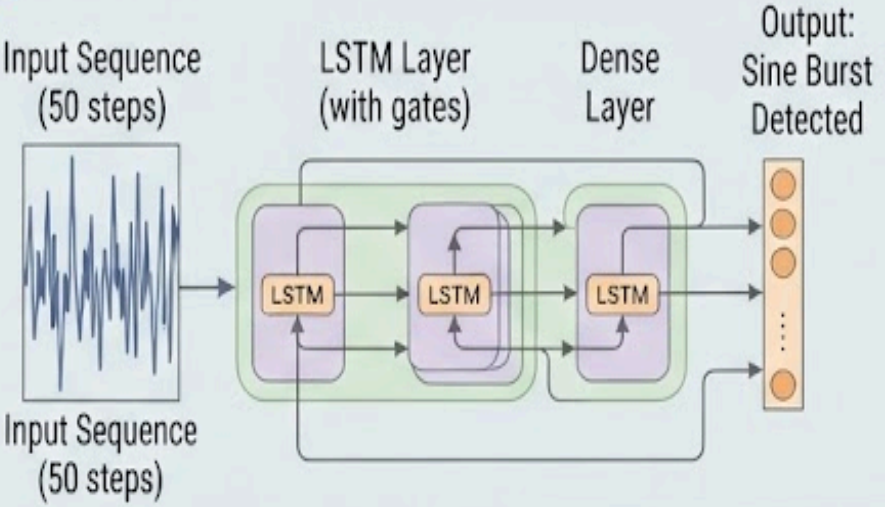


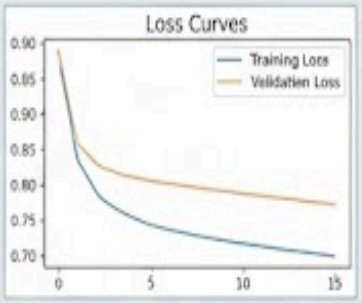


Test Accuracy: 98.00%

Loss curves are in the notebook, with both models trained on the same data

Parrot LSTM





Test Accuracy: 48.50%

Loss curves are in the notebook, with both models trained on the same data

What we learned, who did what, and where to find the full submission.

Reflection: fit the architecture to the task

Similarities: every animal is a network of neurons - inputs are weighted, a bias is added, and an activation fires. Layers turn raw input into a decision.

Differences: how information moves. FNN/CNN move straight ahead. RNN/LSTM carry state step by step. Autoencoders squeeze and rebuild. GNNs pass messages across a web. Transformers let every token attend to every other.

Fit: our test proves the point. A short sine burst in 50 noisy steps favours the Leopard CNN (98.00% test accuracy) over the Parrot LSTM (48.50%), because the burst is local. Change the task to one that depends on distant context, and memory / attention wins. No single animal is best for everything.

Team contributions

Michael Rivas - Notebook Author, Layout & Submission Lead
Wrote the integrated notebook, compiled the team's findings, designed the layout and zoo tour, and handled submission.

Cherluk - Data Engineering & Preprocessing Lead

Olu - Neural Architecture & Computational Demonstration Lead

Taki - Interactive Exhibits, Gallery & Evaluation Lead

Full individual write-ups are in the repo as separate PDFs.

Submission

Git repo (PDF files only): https://github.com/mrivassnj-svg/A03_Team_3_ITAI2376
Start with: A03_Team_3_Index.pdf, then Rivas_A03_Finalized.pdf for Michael's contribution.
Notebook in Canvas: A03_Team_3_Rivas_ITAI2376.ipynb
Deck in Canvas / repo: A03_Team_3_Deck.pdf
Video already submitted in Canvas: A03_Team_3_Rivas_ITAI2376.mp4

Discussion question: which animal would you pick for a task where the answer depends on something said 40 steps earlier - and why?